

WATER SPORTS & RENTAL MANAGEMENT SYSTEM

A SMART DIGITAL SOLUTION FOR
WATER SPORTS ENTHUSIASTS



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THE PLAN

—
ASPECTS TO BE COVERED

01 Sprint Backlog

02 Object Sequence Diagram

03 Class Diagram

INTRODUCTION



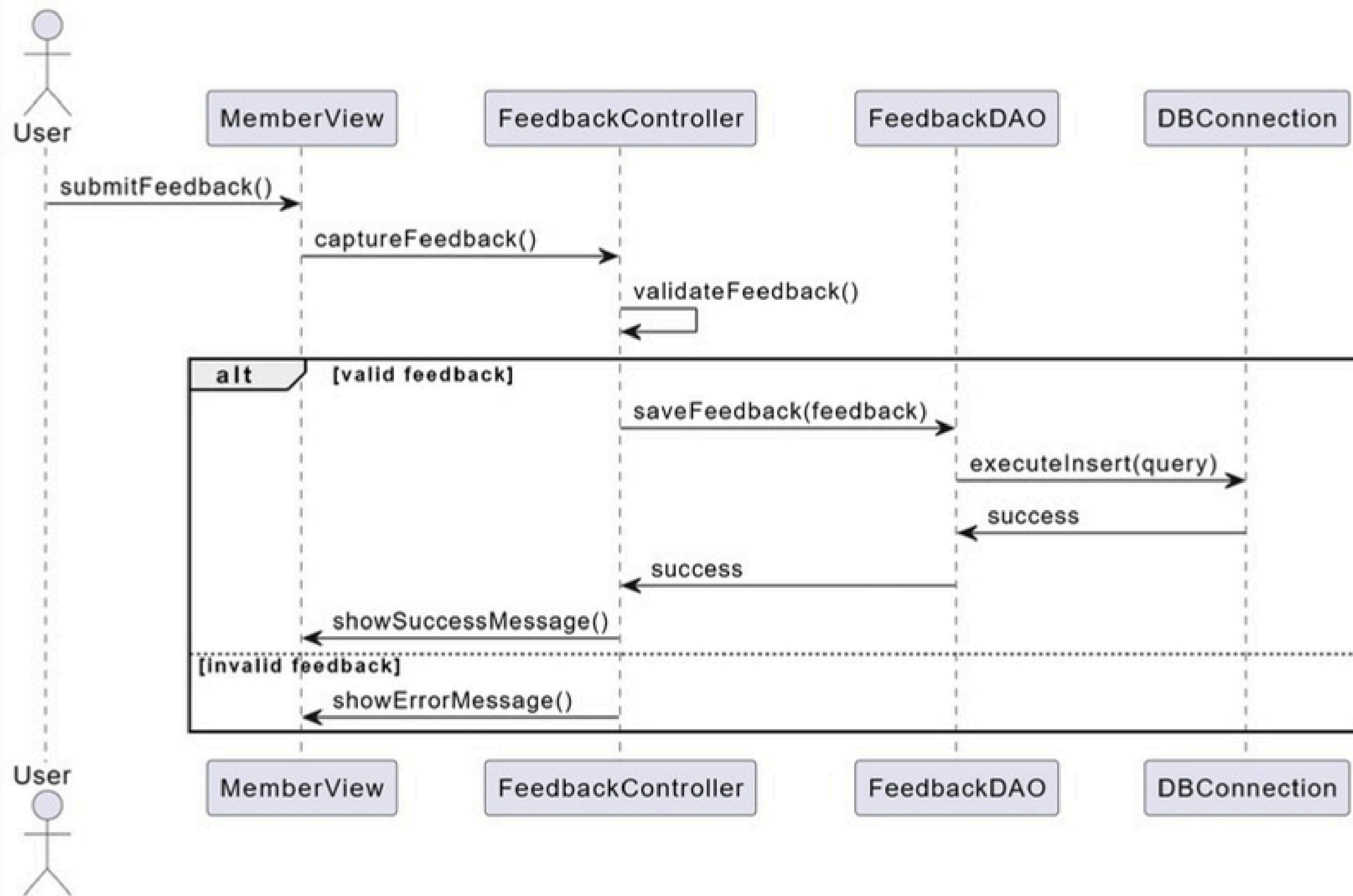
This platform provides a digital solution for managing water sports activities, rentals, and event bookings, ensuring a seamless experience for users.

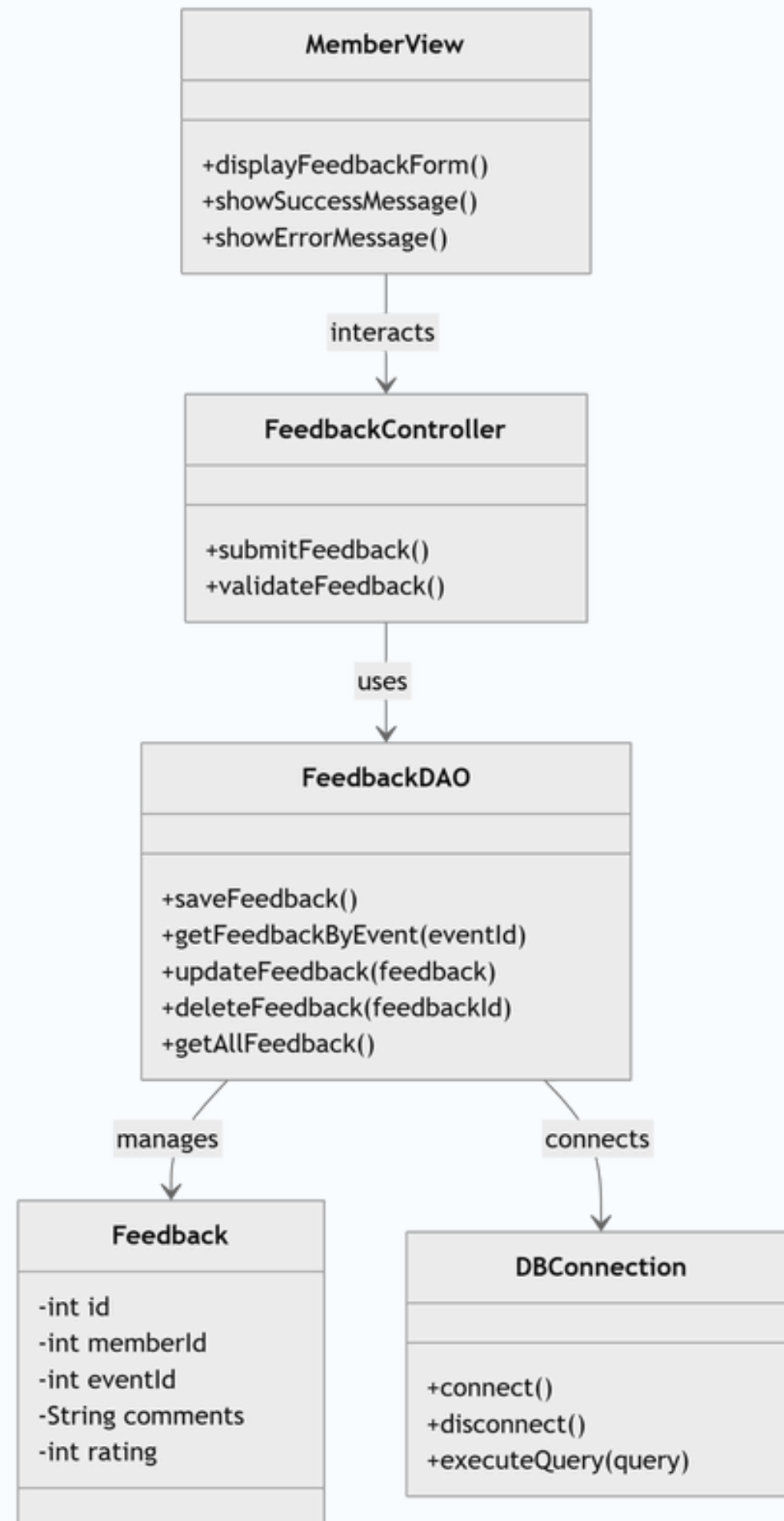
SPRINT 1 BACKLOG

IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
1	AS A MEMBER, I CAN MAKE A PAYMENT FOR AN EVENT AND RECEIVE A TICKET	1.1	DESIGN THE CLASS DIAGRAM FOR PAYMENT & TICKET	1 HOUR	OMAR MZOUGHUI
		1.2	IMPLEMENT ADDPAYMENT() AND GETALLPAYMENTS() METHODS	1 HOUR	
		1.3	IMPLEMENT ADDTICKET() AND GETALLTICKETS() METHODS	1 HOUR	
		1.4	DESIGN AND INTEGRATE THE UI FOR PAYMENT PROCESSING	1 HOUR	
		1.5	IMPLEMENT API INTEGRATION FOR SECURE PAYMENT TRANSACTIONS	2.5 HOURS	
		1.6	HANDLE API RESPONSES TO CONFIRM SUCCESSFUL/FAILED PAYMENTS	2.5 HOURS	

SPRINT 1 BACKLOG

IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
1	AS A MEMBER, I CAN SUBMIT FEEDBACK AFTER ATTENDING AN EVENT	2.1	DESIGN THE FEEDBACK ENTITY AND DATABASE TABLE	1 HOUR	OMAR MZOUGHUI
		2.2	IMPLEMENT ADDFEEDBACK() AND GETALLFEEDBACK() METHODS	1.5 HOUR	
		2.3	CREATE A UI FORM FOR SUBMITTING FEEDBACK	1.5 HOUR	
		2.4	DISPLAY SUBMITTED FEEDBACK IN THE SYSTEM	1 HOUR	





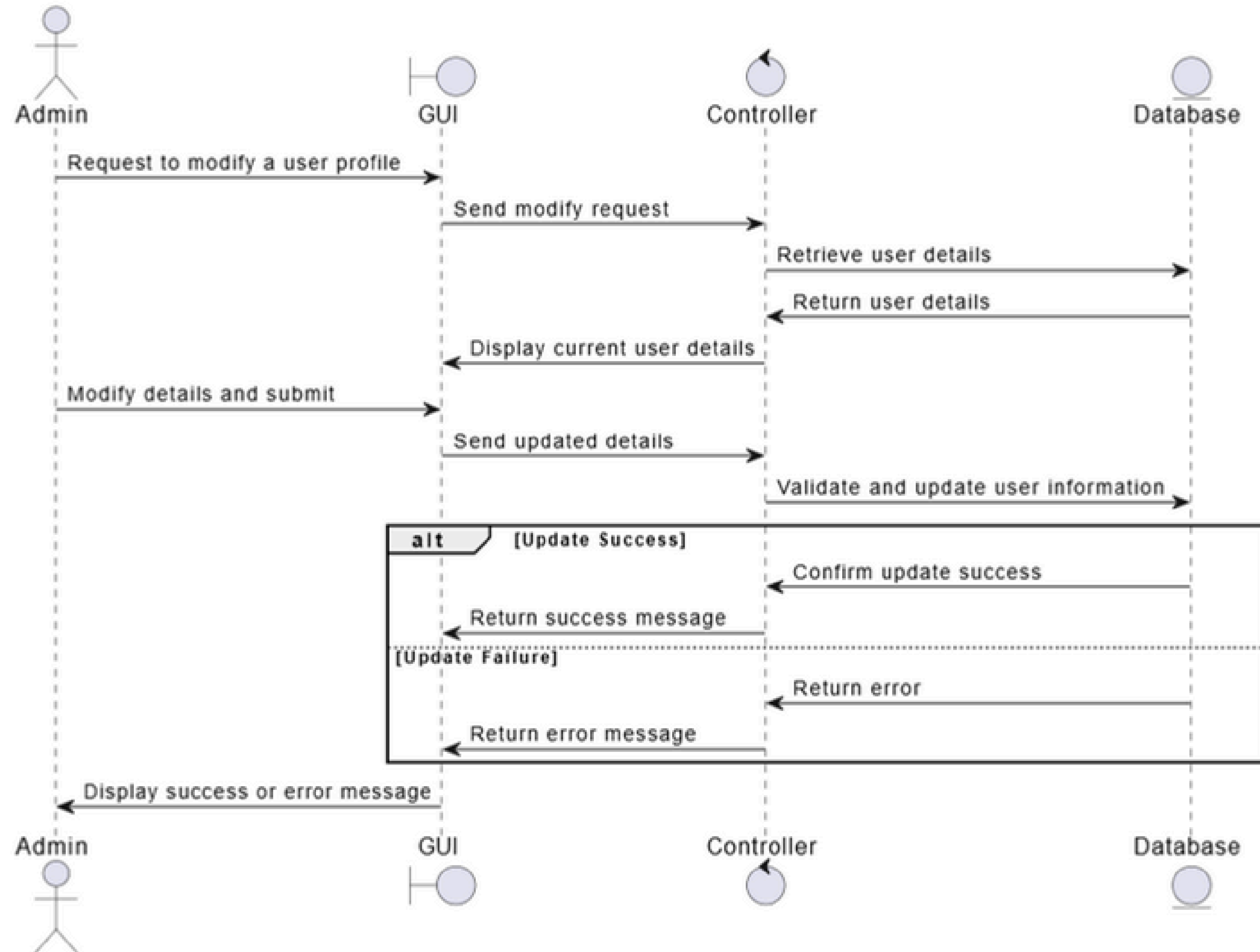
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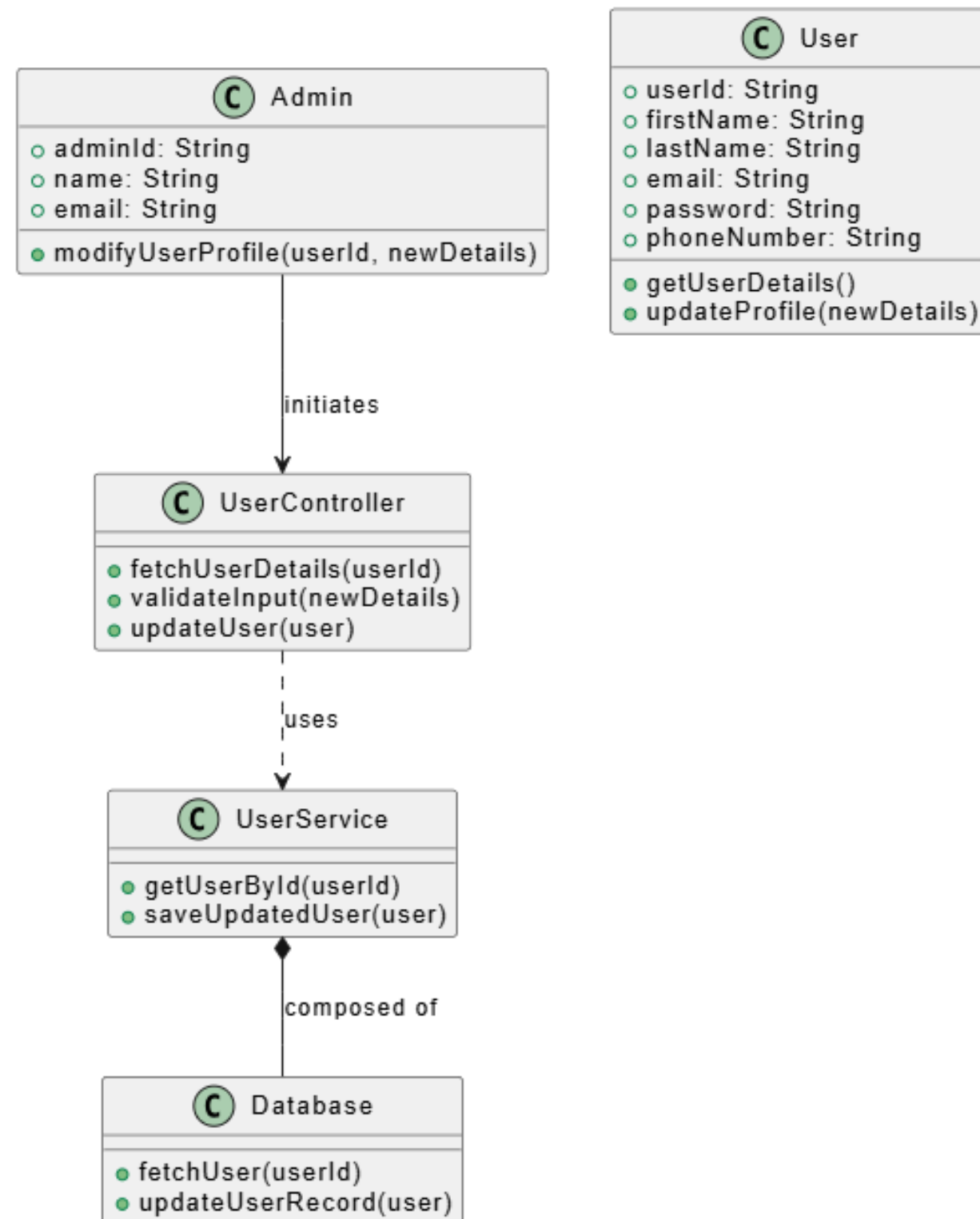
IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
1	AS AN ADMIN, I WANT TO MODIFY A USER'S PROFILE SO THAT I CAN UPDATE THEIR INFORMATION.	1.1	DESIGN THE EDIT USER PROFILE INTERFACE.		3
		1.2	IMPLEMENT THE MODIFYUSER METHOD TO UPDATE USER DETAILS WITH INPUT VALIDATION.		3
		1.3	ALLOW THE ADMIN TO RESET THE USER'S PASSWORD.		3
		1.4	SEND AN EMAIL TO THE USER IF THEIR PROFILE IS MODIFIED.		3
		1.5	TEST THE MODIFYUSER METHOD WITH DIFFERENT SCENARIOS.		3

SPRINT 1 BACKLOG

IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
2	AS A USER, I WANT TO VERIFY MY IDENTITY THROUGH EMAIL AFTER REGISTRATION	2.1	DESIGN THE AUTHENTICATION BUTTON AND USER VERIFICATION FLOW.	3	
		2.2	DEVELOP METHODS TO STORE AND VERIFY USER LOGIN CREDENTIALS.	3	
		2.3	REDIRECT USERS TO THE HOMEPAGE UPON SUCCESSFUL EMAIL VERIFICATION AND LOGIN.	3	
		2.4	SELECT AN AUTHENTICATION API FOR EMAIL VERIFICATION.	3	
		2.5	INTEGRATE THE AUTHENTICATOR TO HANDLE VALID AND INVALID INPUTS CORRECTLY.	3	
		2.6	TEST THE AUTHENTICATION SYSTEM FOR VALID AND INVALID USER INPUTS.	3	

OBJECT SEQUENCE FOR USER MODIFICATION



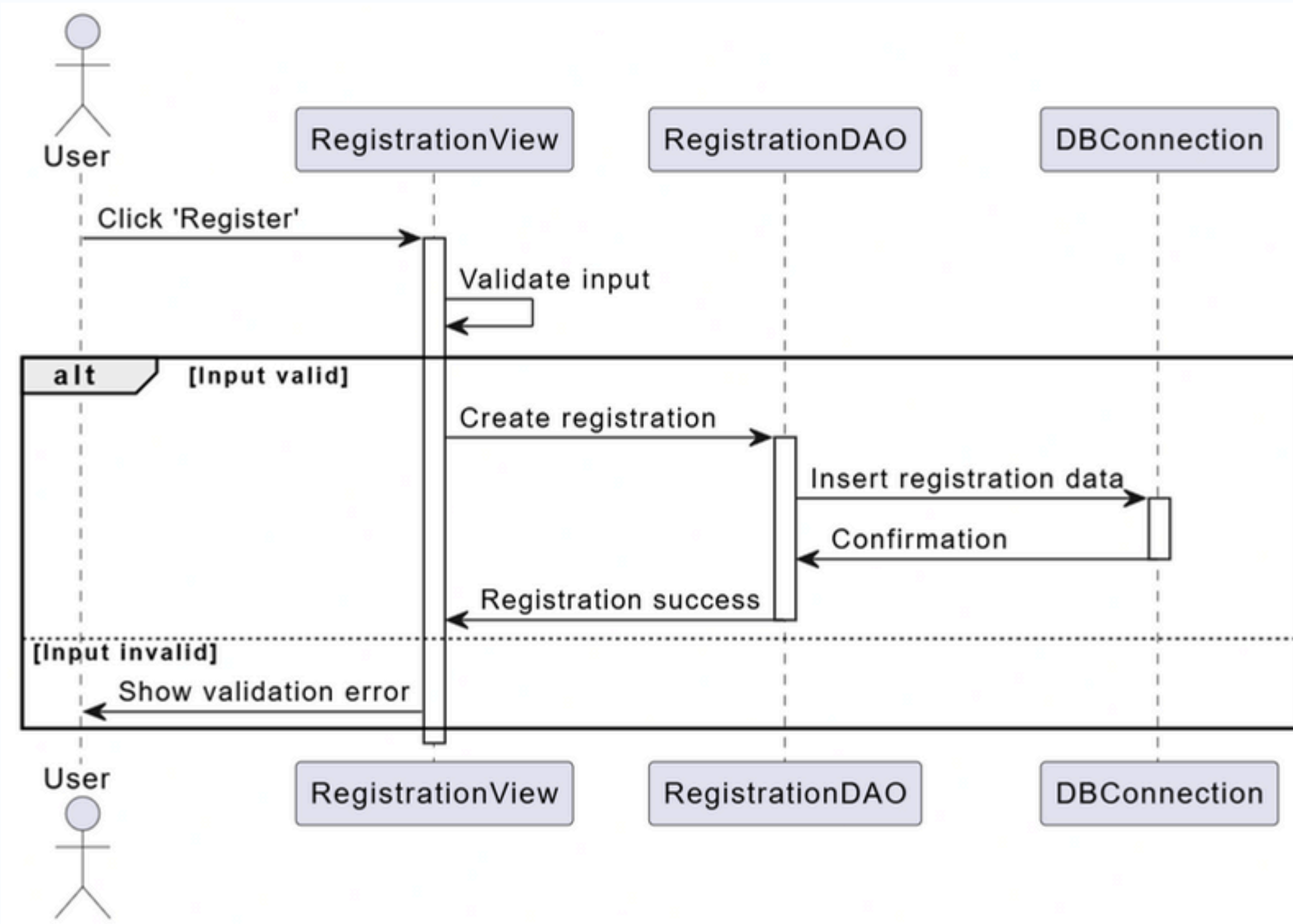


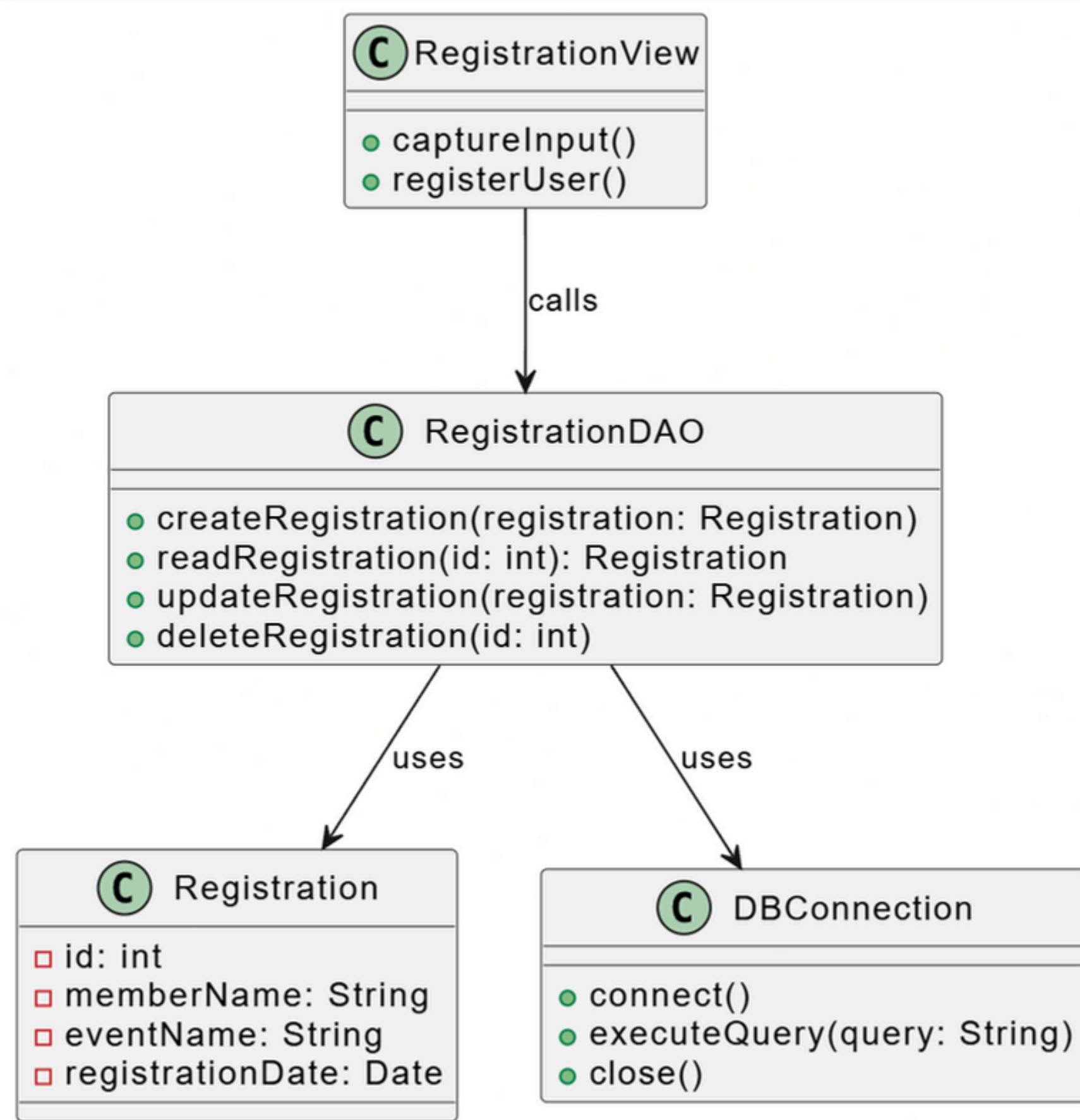
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IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
6	AS A MEMBER, I WANT TO BROWSE AND REGISTER FOR AVAILABLE EVENTS, RECEIVE EMAIL CONFIRMATIONS, AND VIEW MY REGISTRATION HISTORY.	6.1	IMPLEMENT A USER-FRIENDLY EVENT BROWSING INTERFACE WITH SEARCH AND FILTER OPTIONS.	2	
		6.2	DEVELOP A REGISTRATION METHOD THAT VERIFIES EVENT AVAILABILITY AND USER STATUS.	1.5	
		6.3	INTEGRATE AN EMAIL SERVICE (E.G., JAVAMAIL API) TO SEND REGISTRATION CONFIRMATIONS.	2	
		6.4	CREATE A "MY EVENTS" SECTION WHERE USERS CAN VIEW PAST AND UPCOMING REGISTRATIONS.	1.5	
		6.5	IMPLEMENT CANCELLATION FUNCTIONALITY WITH CONFIRMATION PROMPTS AND EMAIL NOTIFICATIONS.	1.5	
		6.6	TEST THE REGISTRATION FLOW, INCLUDING EDGE CASES (E.G., DOUBLE REGISTRATION, EVENT FULL).	1.5	

SPRINT 1 BACKLOG

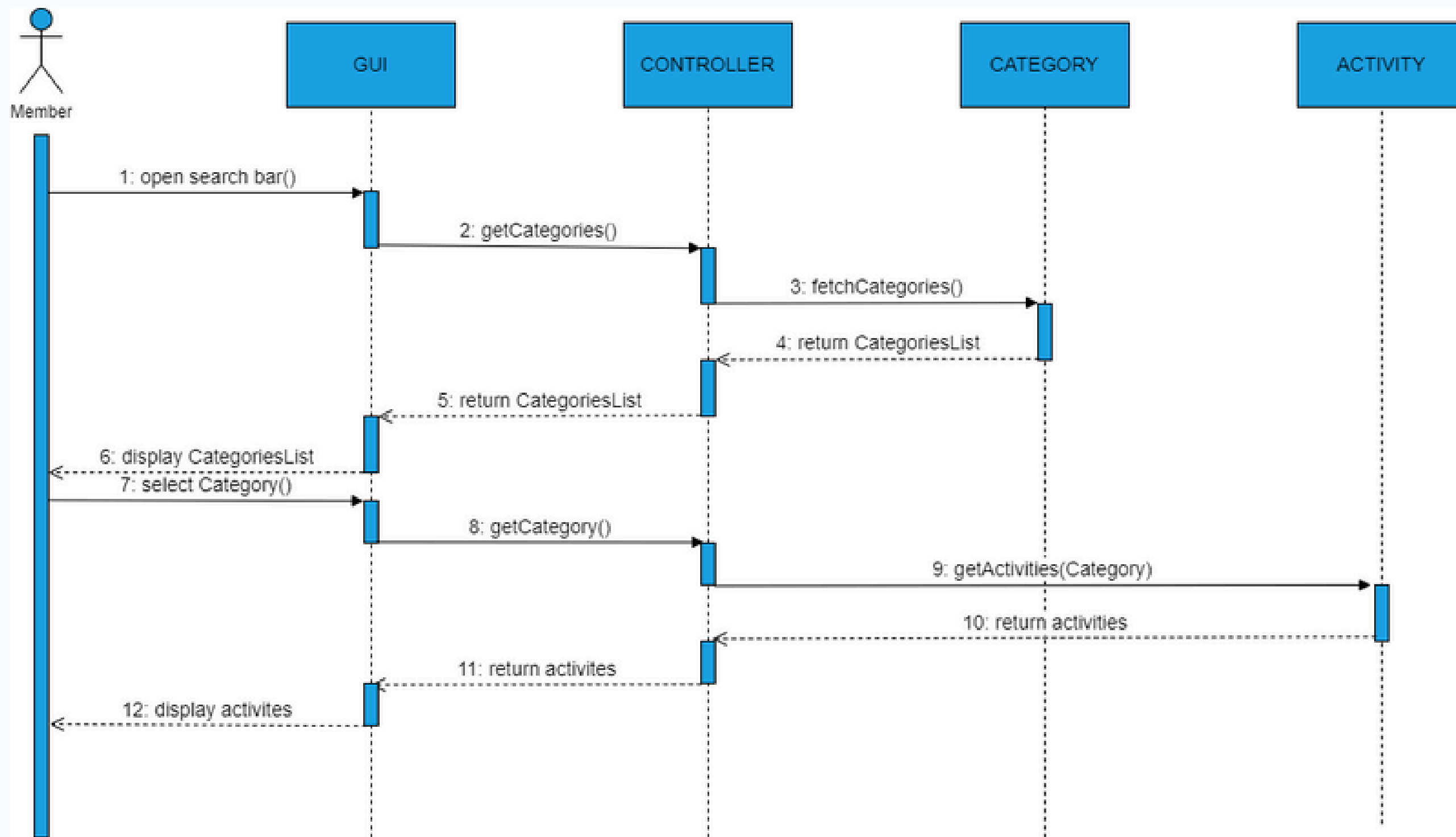
IDF	USER STORY	TASK ID	TASK	ESTIMATION	RESPONSIBLE
5	AS AN ADMIN, I WANT TO INTEGRATE AN EXTERNAL API TO FETCH EVENT DATA AND SYNCHRONIZE IT WITH OUR LOCAL DATABASE.	5.1	RESEARCH AND SELECT A SUITABLE EVENT API (E.G., EVENTBRITE, MEETUP).	1.5	
		5.2	IMPLEMENT AN EVENTAPISERVICE CLASS TO HANDLE API REQUESTS AND PARSE RESPONSES.	2	
		5.3	CREATE A SYNCEVENTS METHOD IN EVENTDAO TO MERGE EXTERNAL EVENT DATA WITH THE LOCAL MYSQL DATABASE.	2	
		5.4	ADD A "SYNC WITH API" BUTTON IN THE EVENT MANAGEMENT UI.	1	
		5.5	IMPLEMENT ERROR HANDLING AND LOGGING FOR API REQUESTS (E.G., HANDLING RATE LIMITS, INVALID RESPONSES).	1.5	
		5.6	TEST THE API INTEGRATION, INCLUDING SCENARIOS WITH NEW, UPDATED, AND DELETED EVENTS.	1.5	

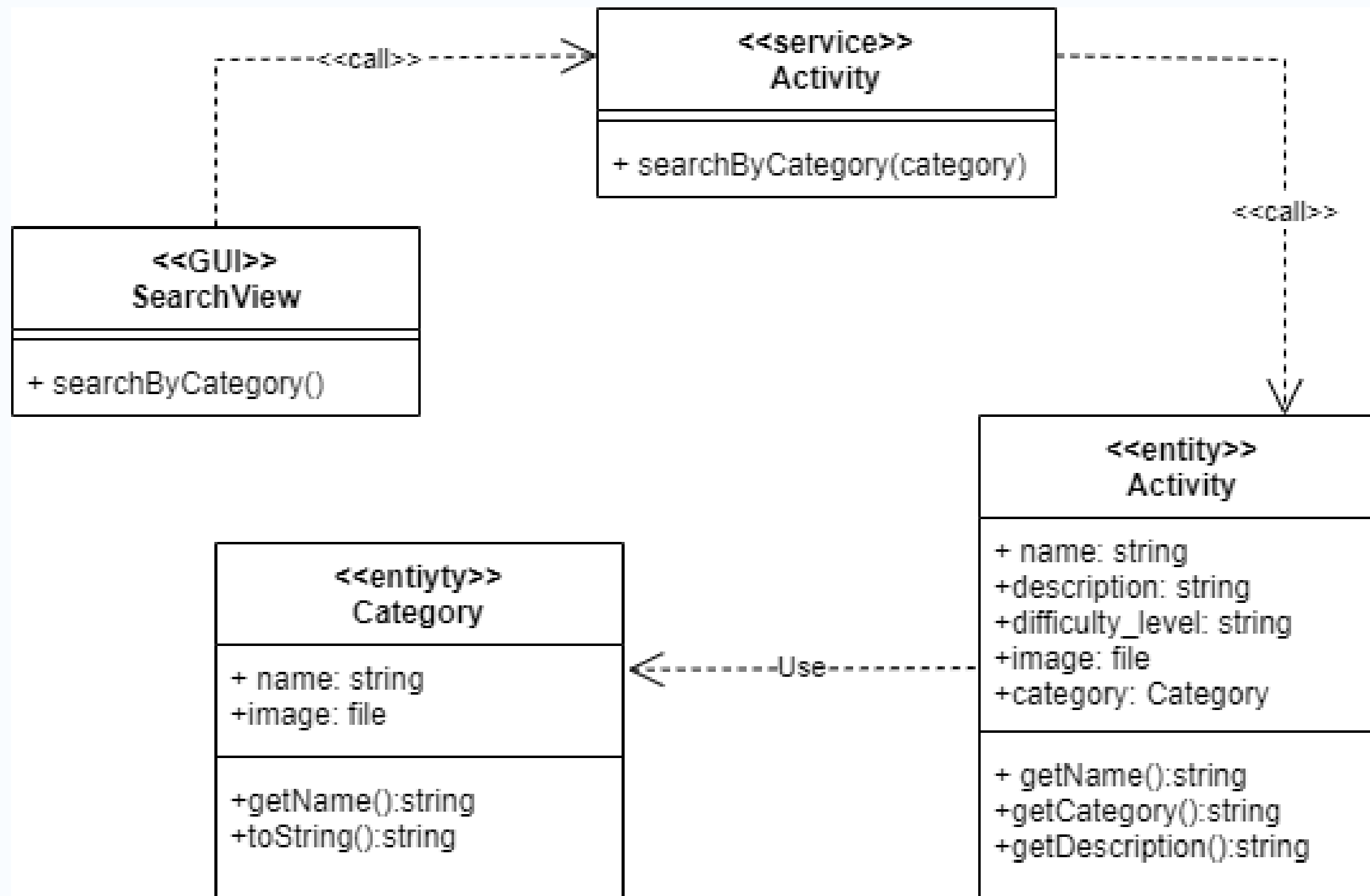




IDF	USER STORY	TASK ID	TASK	ESTIMATION
1	AS A MEMBER I WANT TO FILTER ACTIVITIES BASED ON CATEGORY	1.1	DESIGN THE RELATIVE CLASS DIAGRAM	1 HOUR
		1.2	DESIGN THE ACTIVITY INTERFACE	1 HOUR
		1.3	CREATE METHOD FILTER CATEGORIES	2 HOUR
		1.4	CREATE CATEGORY.SQL TABLE	1 HOUR

IDF	USER STORY	TASK ID	TASK	ESTIMATION
1	AS A MEMBER I WANT TO REGISTER FOR AN ACTIVITY	1.1	DESIGN THE RELATIVE CLASS DIAGRAM	1 HOUR
		1.2	DESIGN THE REGISTRATION FORM INTERFACE	1 HOUR
		1.3	CREATE THE METHOD ACTIVITY REGISTER	2 HOUR
		1.4	CREATE THE ACTIVITY REGISTRATION TABLE	1 HOUR





MODULE: RENTAL EQUIPMENT AND HISTORY

MOHAMED AMINE IBRAHIM

SPRINT 1 BACKLOG

IDF	USER STORY	TASK ID	TASK	ESTIMATION
1	AS A MEMBER, I WANT TO BROWSE AVAILABLE RENTAL EQUIPMENT AND FILTER BY AVAILABILITY AND TYPE SO THAT I CAN EASILY FIND WHAT I NEED.	1.1	DESIGN THE CLASS DIAGRAM FOR BROWSING AND FILTERING RENTAL EQUIPMENT.	
		1.2	IMPLEMENT THE RENTAL CLASS WITH FILTERING LOGIC.	
		1.3	DEVELOP THE UI FOR BROWSING USING JAVA FX.	
		1.4	IMPLEMENT DAO TO RETRIEVE EQUIPMENT FROM THE DATABASE.	

IDF	USER STORY	TASK ID	TASK	ESTIMATION
	AS A MEMBER, I WANT TO CHECK REAL-TIME AVAILABILITY OF RENTAL EQUIPMENT USING AN EXTERNAL API SO THAT I CAN ENSURE MY BOOKING IS POSSIBLE.	2.1	RESEARCH AND INTEGRATE THE AVAILABILITY API.	
		2.2	IMPLEMENT AVAILABILITYSERVICE.JAVA TO HANDLE API CALLS.	
		2.3	HANDLE API RESPONSES AND UPDATE THE UI.	
		2.4	IMPLEMENT UNIT TESTS FOR API INTEGRATION.	

